Samir Ghosh

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Education

University of California, Santa Cruz — Santa Cruz, California
PhD Candidate in Computational Media · 2022 – *Present*
Advisor: Katherine Isbister
Expected graduation: December 2026

University of Southern California — Los Angeles, California
BS in Computational Linguistics · 2018
BA in Cognitive Science

Selected Projects

Virtual Reality for Scientific Sensemaking
Multi-year research program in collaboration with civil engineers, marine scientists, and other technical domains to explore how collaborative XR tools can augment the scientific sensemaking process. Supported by the Sloan Foundation.

AI War Cloud
Ars Electronica 2025 piece by Sarah Ciston that illustrates complex relations of AI technology across industry and government. Developed XR views using a bespoke rendering technique. Winner of the S+T+ARTS Grand Prize of the European Commission.

Booksnake AR
NEH funded iOS AR app displaying assets from the Library of Congress and other archival information. Early design, software architecture, and project management contributions.

Bunker Hill VR
Historical recreation of 1930s downtown Los Angeles using civil engineering data. Project management, documentation, and technical contributions.

Bodyscape
Internationally exhibited (e.g. SIGGRAPH, Ars Electronica) wearable technology fashion piece by Behnaz Farahi. Contributions in gait responsive algorithms, circuit design, safety systems, and generative design.

Honors & Scholarships

Dean's Travel Grant	2025
Finalist, UC Santa Cruz Grad Slam	2023
Regents Fellowship	2023
CITRIS Tech For Good Award	2022
Dean's Fellowship	2022

Publications

AI WAR CLOUD DATABASE

Samir Ghosh and Sarah Ciston

SIGGRAPH Spatial Storytelling '26

Designing Shared Virtual Reality Sensemaking Tools for Wildfire Mitigation DOI pending

Samir Ghosh, Maryann Godje, Charles Lesser, Yanglan Wang, Paola Lorusso, Kenichi Soga, Joshua McVeigh-Schultz, and Katherine Isbister

Meaningful XR '26

Diving in a Coral Reef: Collaborative XR Exploration of Reef Restoration using Gaussian Splats DOI pending

Samir Ghosh, Erik Chao, Christopher Lowrie, Will Gislason, Michael Beck, Joshua McVeigh-Schultz, and Katherine Isbister

Meaningful XR '26 Demo

“Let's Jump into More Creative Avatars and Take this Brainstorm to the Flying Platform:” Playful Prototypes of VR Meeting Support Tools

Anya Osborne, Joshua McVeigh-Schultz, Alexandra Leeds, George Butler, **Samir Ghosh**, and Katherine Isbister

DIS 2025

Introducing Booksnake: A Scholarly App for Transforming Existing Digitized Archival Materials into Life-Size Virtual Objects for Embodied Interaction in Physical Space, using IIIF and Augmented Reality

Sean Fraga, Christy Ye, Henry Huang, Zack Sai, Michael Hughes, April Yao, and **Samir Ghosh**

CHI EA '25

Purposeful XR: Affordances, Challenges, and Speculations for an Ethical Future

Elizabeth Childs, **Samir Ghosh**, Sebastian Cmentowski, Andrea Cuadra, Rabindra Ratan

CHI EA '25

Designing Shared VR Tools for Spatial Scientific Sensemaking About Wildfire Evacuation

Samir Ghosh, Yuhui Wang, William Zhou, Kelly Lin, Joshua McVeigh-Schultz, and Katherine Isbister.

CHI EA '24

Eye Ball: Gazing as a Dilemma in a Competitive Virtual Reality Game

Michael Lankes, **Samir Ghosh**, Charles Bishop Lesser, and Katherine Isbister.

CHI EA '24

The Cuteness Factor: An Interpretive Framework for Artists, Designers and Engineers

Angela Y.H. Fan, Chen Ji, Ella Dagan, **Samir Ghosh**, Yuhui Wang, Katherine Isbister.

DIS 2023

Workshop Papers

Brahma: an Open Source Library for Cross-platform Social XR and Scientific Visualization
Samir Ghosh, Joshua McVeigh-Schultz, and Katherine Isbister
Social Augmentation through XR Technologies Workshop, CHI '26

Designing Interaction Approaches for Shared Sensemaking in XR
Samir Ghosh, Joshua McVeigh-Schultz, and Katherine Isbister.
Sensemaking Workshop, CHI '24

Social Physiological Data Awareness in Collocated Mixed Reality Movement
Samir Ghosh, Charles Lesser, Kaia Rae Schweig, Sofia De La Vega Mireles, and Katherine Isbister.
PhysioCHI Workshop, CHI '24

Revealing Aspects of Hawai'i Tourism Using Situated Augmented Reality
Karen Abe, Jules Park, and **Samir Ghosh**
With or Without Permission Workshop, CHI '24

Designing a mixed-initiative multi-user VR interface for wildfire mitigation
Samir Ghosh, Yanglan Wang, Kecheng Cheng, Anthony Angeles, Andrew Moskovich, Kenichi Soga, and Katherine Isbister.
HCI for Climate Change Workshop, CHI 2023

Research Positions

Graduate Student Researcher, UC Santa Cruz — Jun 2025 – Present
Coastal Climate Resilience Center and UC Santa Cruz Visualization Lab

Graduate Student Researcher, UC Santa Cruz — Jun 2025 – Present
Catalyst Program, UC Open Source Program Office

Visiting Scholar, Blekinge Institute of Technology — May 2025 – Jun 2025
HINTS Project, Department of Computer Science

Graduate Student Researcher, UC Santa Cruz — Jun 2023 – Apr 2025
Virtual Reality for Scientific Sensemaking, Social Emotional Technology Lab
Supported by the Sloan Foundation

Industry Experience

Ahmanson Lab — Los Angeles, California
USC Harman Academy
Assistant Director · Jan 2019 – Aug 2022

YUR Inc. — Los Angeles, California
VR Developer · Jul 2021 – Dec 2021

Intel Corporation — Santa Clara, California
Intern · Summer 2016, Summer 2018, Fall 2018

Enlighted Inc. — Sunnyvale, California
Intern · Summer 2014

Teaching

Teaching Assistant, UC Santa Cruz — Winter 2022
CMPM 115: Lead By Design

Seminars, Talks & Tutorials

Seals, Proteins, and Wildfires — May 2026
Lightning Talk, Questioning Reality '26 (QR'26)

A Career in VR + Coding and Art Workshop — Jan 2026
Bullis Charter School, Los Altos, California

Virtual Reality and Scientific Sensemaking — Jun 2025
Department of Industrial Design, TU Eindhoven, Netherlands

Virtual Reality and Scientific Sensemaking — Jun 2025
German Aerospace Agency (DLR), Braunschweig, Germany

Virtual Reality and Scientific Sensemaking — May 2025
Department of Computer Science, Blekinge Institute of Technology, Sweden

Revisiting the for Loop — Nov 2023
Slugworks, UC Santa Cruz

A Career in HCI and VR — Oct 2023
Cognitive Science Student Association, UC Santa Cruz

Generative Art in Virtual Reality Using p5js — Jun 2023
Digital Arts and New Media, UC Santa Cruz

Wildfires in Virtual Reality — Mar 2023
UC Santa Cruz Grad Slam, Kuumbwa Jazz Center

Multi-user VR Workshop — Feb 2023
Digital Arts and New Media, UC Santa Cruz

Surveillance and the Attention Economy — Mar 2022
Polymathic Making Workshops, Ahmanson Lab at USC

Computational Art — Feb 2022
Polymathic Making Workshops, Ahmanson Lab at USC

WebRTC, WebGL, and other web protocols — Jan 2022
Polymathic Making Workshops, Ahmanson Lab at USC

Techniques with Graphics Code — Sep – Oct 2021
Emergent Technology Series, Ahmanson Lab at USC

Sensors, Lights, and Motors — Sep 2021
Polymathic Making Workshops, Ahmanson Lab at USC

3D Modeling Basics — Sep 2021
Polymathic Making Workshops, Ahmanson Lab at USC

Glitch + D3.js — Oct 2020
Generative art-a-thons, Ahmanson Lab at USC

VR with Mozilla Hubs — Oct 2020
Generative art-a-thons, Ahmanson Lab at USC

p5js — Sep 2020
Generative art-a-thons, Ahmanson Lab at USC

Applied Neural Networks — Apr 2020
Polymathic Making Workshops, Ahmanson Lab at USC

Introduction to Creative Code — Apr 2020
Polymathic Making Workshops, Ahmanson Lab at USC

STEM Speaker Series — Mar 2020
Katherine Johnson STEM Academy

Deepfake Detection — Mar 2020
Polymathic Making Workshops, Ahmanson Lab at USC

Practical Arduino — Mar 2020
Polymathic Making Workshops, Ahmanson Lab at USC

Data Surveillance and Digital Rights — Feb 2020
Polymathic Making Workshops, Ahmanson Lab at USC

Introduction to 3D Printing and the Makerbot Replicator — Feb 2020
Polymathic Making Workshops, Ahmanson Lab at USC

WebVR — Nov 2019
Polymathic Making Workshops, Ahmanson Lab at USC

Wearable Technology — Oct 2019
Polymathic Making Workshops, Ahmanson Lab at USC

Get Your Own Climate Data — Oct 2019
Polymathic Making Workshops, Ahmanson Lab at USC

VR Web Development — Sep – Oct 2021
Emergent Technology Series, Ahmanson Lab at USC

Object Recognition, Privacy Rights, and Data Collection — Sep 2021
Polymathic Making Workshops, Ahmanson Lab at USC

Introduction to 3D Printing and the Makerbot Replicator — Sep 2019
Polymathic Making Workshops, Ahmanson Lab at USC

Practical Arduino — Sep 2019
Polymathic Making Workshops, Ahmanson Lab at USC

Promise and Peril of Algorithmic Living — Apr 2018
USC Visions and Voices

Service

Google Summer of Code — Summer 2025 – Present
Mentor · *UC Open Source Program Office*

Creative Code Collective — 2023 – Present
Community Lead

Committee for Planning and Budget — 2023 – Present
Graduate Representative · *Academic Senate, UC Santa Cruz*

MIT Reality Hack — 2025
Mentor

ACM Halfway to the Future Symposium 2024 — 2023 – 2024
Social Media Chair

Google Summer of Code — Summer 2022
Mentee, Contributor · *Processing Foundation*

Corpus Callosum (Student Org.) — Fall 2015 – Spring 2018
Technical Director · *Viterbi School of Engineering, USC*

Personal Information

Citizenship	USA
Languages	English (native), French (proficient), Korean (basic)
Email	samir.ghosh@ucsc.edu
Website / Portfolio	samir.tech
Interests	Dance, open water swimming, creative code